PerturbCausal: Virtual Cell Model Predictions Do Not Substitute for Real Perturb-seq in Causal Network Recovery

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01 Background

VCMs promise in-silico CRISPR screens

- Virtual cell models are marketed as in-silico substitutes for CRISPR screens that cost millions of dollars per cell line.
- If their predictions were trustworthy, much of perturbation biology could move in-silico.

Prior work: they fail at per-gene prediction

- VCMs do not beat simple linear baselines at per-gene expression prediction (Ahlmann-Eltze & Huber 2025, Nat Methods).
- But per-gene accuracy is a property of the marginal distribution only.

Can a cheap virtual cell replace a real screen, downstream?

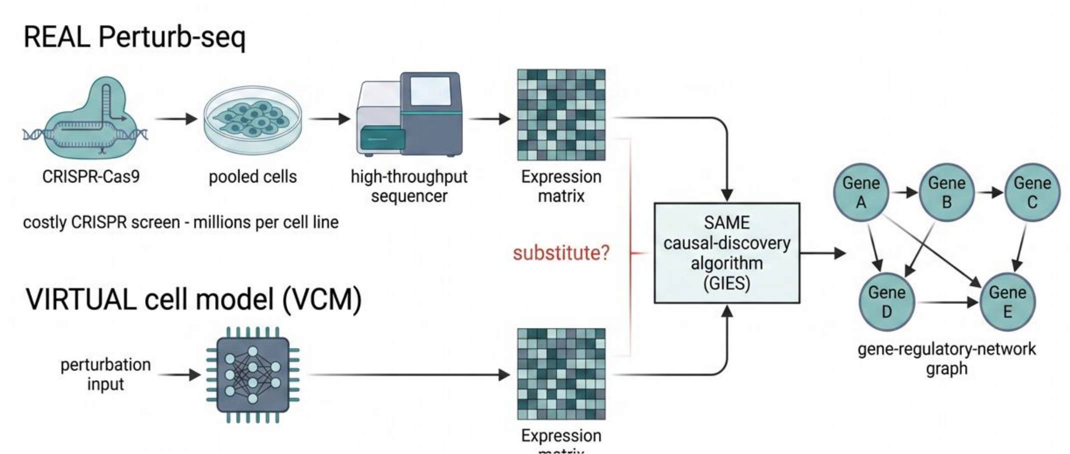


Fig 1. Real and virtual Perturb-seq feed one fixed causal-discovery engine. We ask whether the in-silico lane substitutes for the expensive real lane in the recovered network.

The payoff is downstream, in the joint structure

- Drug-target ID, combination prediction, and GRN inference depend on the joint distribution across genes.

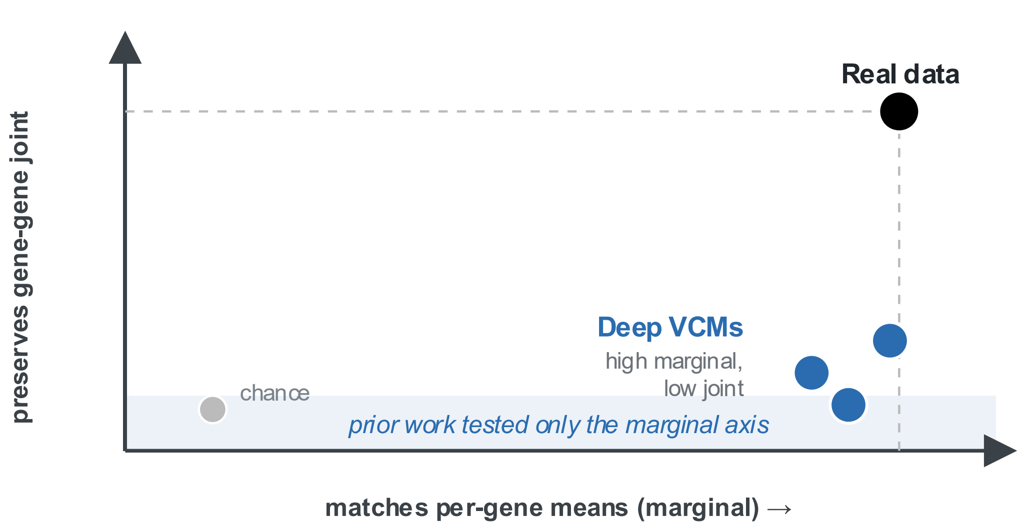


Fig 2. Two different axes. Deep VCMs sit high on the marginal, low on the joint; prior work tested only the marginal axis.

The open question, and our contribution

- Nobody had tested whether predicted data can substitute for real data downstream in causal-network recovery.
- This is the first benchmark that does, holding the causal algorithm fixed and varying only the data source.

Objective: measure how far predicted data can substitute for real data in causal-network recovery, and decompose any gap into a magnitude component and a joint-structure component.

02 Methods

Fixed-algorithm, vary-data-source design

- Metric: edge-set F1 of the predicted-data network against the network the same algorithm recovers from real data.
- Only the data source varies; the discovery algorithm is held fixed.
- Edge-set F1 weighs precision (few spurious edges) against recall (few missed real edges), so a method scores only for recovering the true wiring, not for matching effect counts.

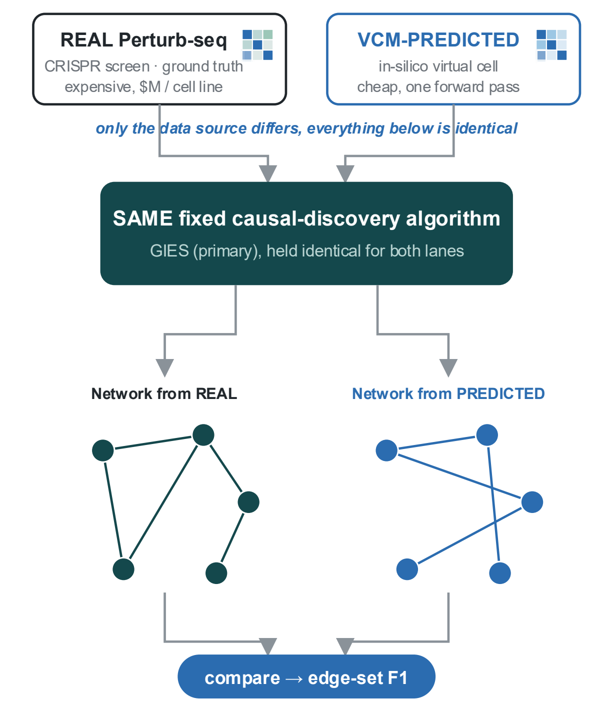


Fig 3. Only the data source varies; the same fixed algorithm (GIES primary) runs on both, and we compare recovered edge sets.

Predictors: four deep VCMs and one linear

GEARS · graph neural network

CPA · conditional VAE

Geneformer · 104M-parameter transformer

STATE · Arc Institute 2025 foundation model

- Comparator: sparse-linear ElasticNet fit on held-out controls, a deliberately simple baseline.

Causal-discovery backends (held fixed)

- GIES: score-based, interventional (primary).
- inspre (sparse inverse of the ACE matrix), PC (constraint-based), and NOTEARS (continuous optimization).

Datasets and chance reference

- Replogle K562 (primary, 200-gene subset), RPE1 (cross-context), and Norman 2019 (cross-paradigm CRISPRa). CRISPRi and CRISPRa.
- Chance is a per-backend permutation null (shuffle each ACE column across perturbations), giving a [2.5, 97.5] band.

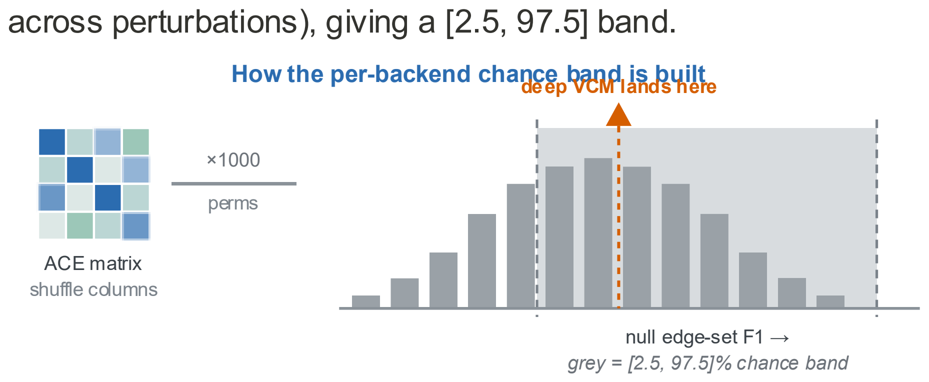


Fig 4. How the per-backend chance band is constructed.

03 Results

Deep VCMs tie a sparse linear baseline

- K562, N=200, 5 seeds: no pairwise difference at $p<0.05$, TOST-equivalent at $\Delta = 0.045$ F1 (CPA +0.014, GEARS +0.020, Geneformer +0.0002).
- A 104M transformer, a GO-prior GNN, and a conditional VAE all tie a linear regression.

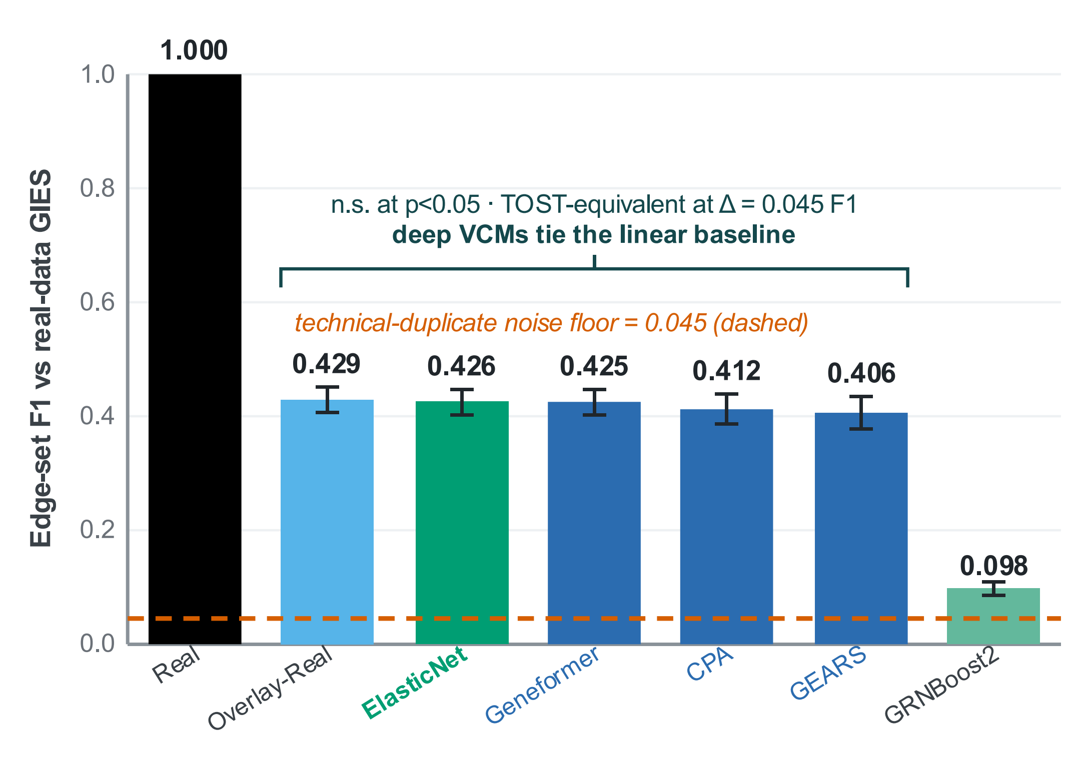


Fig 5. Edge-set F1 vs the real-data GIES network (K562, N=200, 5 seeds); 95% CI whiskers.

Holds across backends and datasets

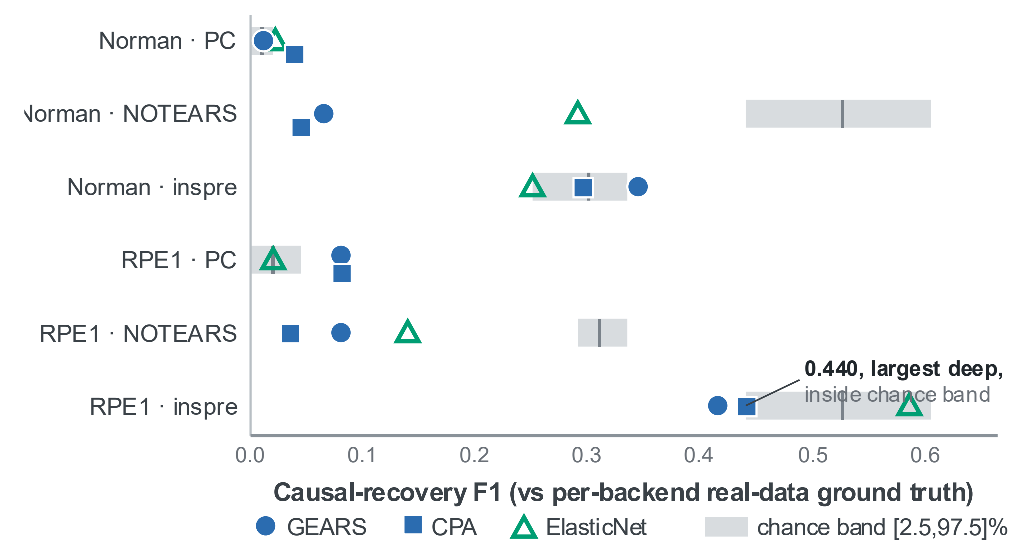


Fig 6. Deep VCMs vs the per-backend chance band [2.5, 97.5]%, across {RPE1, Norman} by {PC, NOTEARS, inspre}.

- Largest deep-model F1 across the six cells is 0.440 (CPA, inspre, RPE1), inside the chance band.
- ElasticNet is strongest in 4 of 6; no deep VCM exceeds it on any backend or dataset.

Mechanism: effect-size inflation is mode collapse

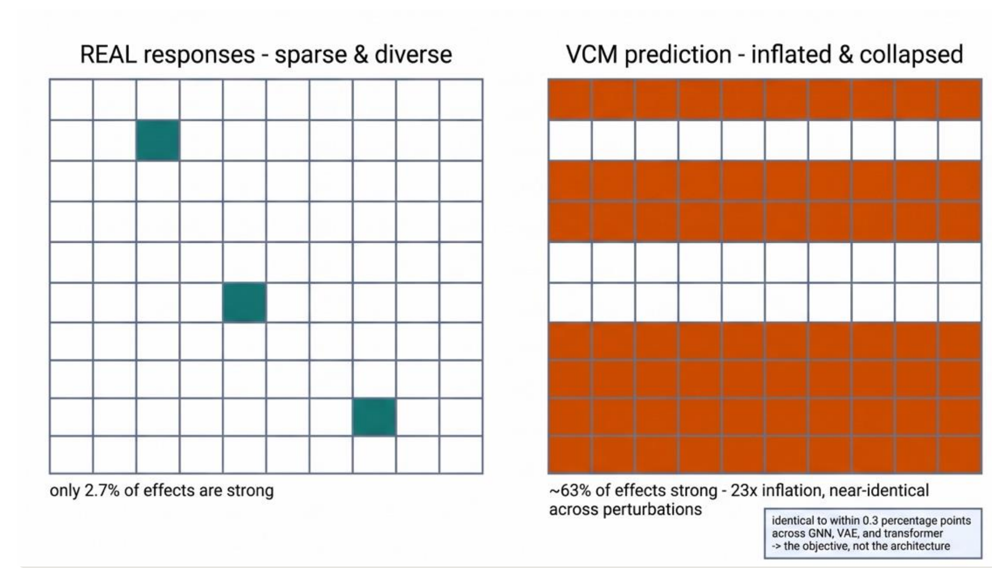


Fig 7. Real cells respond sparsely and diversely; VCMs light up nearly every gene, near-identically across perturbations.

- Real 2.7% of effects exceed |log2FC| = 0.5; deep models inflate to about 63% (GEARS 63.2, CPA 63.4, Geneformer 63.5), a 23x inflation identical to within 0.3 pp across architectures.
- CPA inter-perturbation cosine is 0.84 vs real 0.08. inspre probe: CPA and Geneformer recover ZERO edges where real and ElasticNet recover about 12,000.

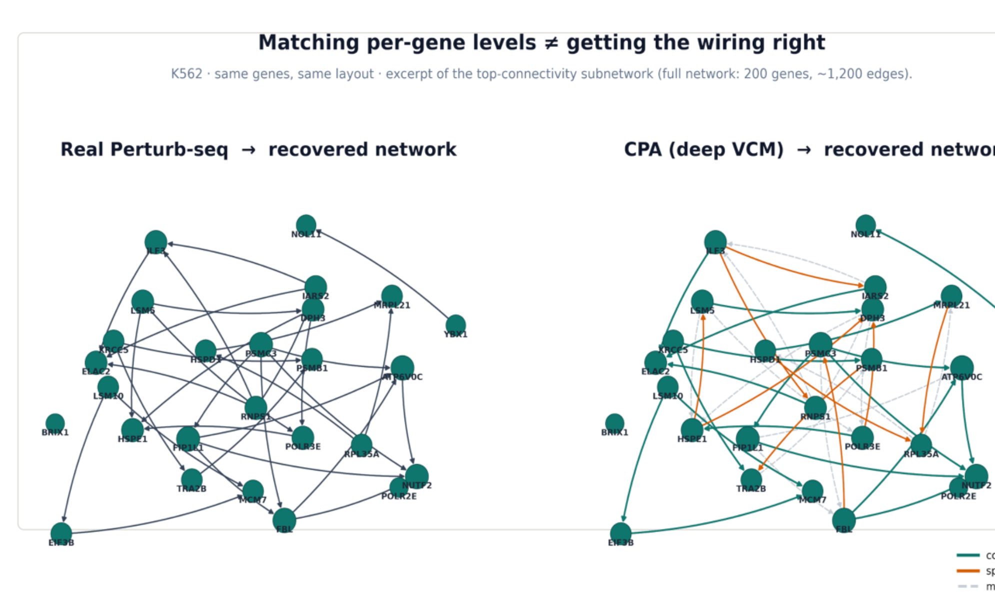
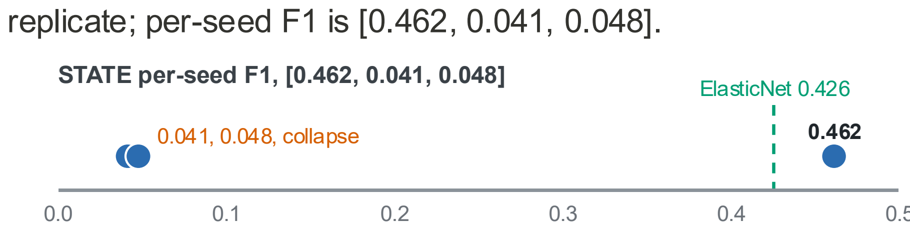


Fig 8. Real K562 recovered network vs CPA (deep VCM), same genes and layout; teal = correct edge, vermillion = spurious, grey dashed = missed. Full-network edge-set F1 = 0.42.

The 2025 foundation model, too

- STATE single-seed 0.462 beats seed-42 ElasticNet but does not replicate; per-seed F1 is [0.462, 0.041, 0.048].



04 Findings & Conclusions

Calibration corrects magnitude, not causality

- Quantile-matching fixes the magnitude marginal exactly (onto real = 0.114), yet held-out 160/40 causal F1 does not close: GEARS 0.303 to 0.290, CPA 0.309 to 0.289, Geneformer 0.319 to 0.287, ending below ElasticNet (0.425).

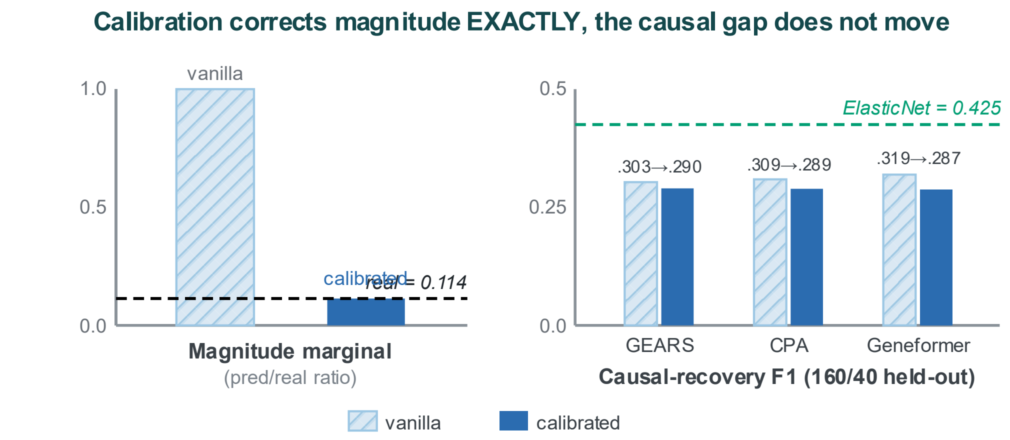


Fig 9. Quantile-matching pins the magnitude marginal onto real (0.114) exactly, yet held-out causal F1 barely moves.

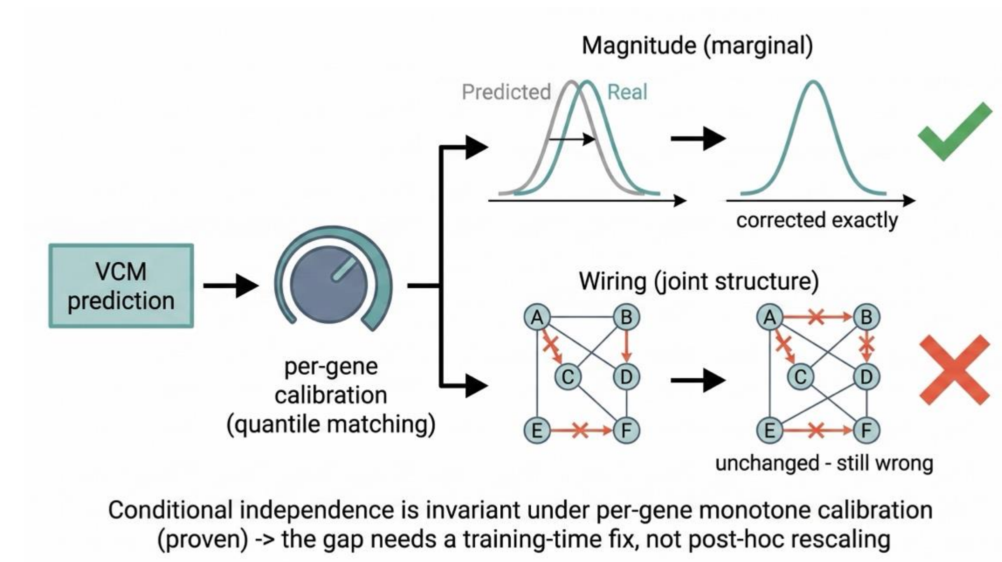


Fig 10. Calibration turns a per-gene dial to fix the ruler; the scrambled wiring underneath is untouched.

Why: an identifiability result

Proposition. Conditional independence is invariant under per-gene monotone (rank-preserving) calibration, so marginal rescaling provably cannot change the recovered graph.

- The residual is the off-diagonal joint (precision-matrix support) structure, so closing the gap needs a training-time intervention, not post-hoc rescaling.

The gap is reachable

- A no-GPU learned joint-transform proxy raises held-out CPA causal F1 from 0.424 to 0.535 (+0.111), the first evidence the gap is reachable by a joint operation.
- A naive covariance-Frobenius penalty alone fails a pre-registered multi-seed gate (+0.003), so it is the learned joint map, not the raw penalty.

Conclusions

- The substitutability gap decomposes into a magnitude marginal (calibration-correctable) plus a joint-distribution residual that calibration cannot reach.
- No deep VCM beats a sparse linear baseline on causal recovery; none reaches real-data quality.
- Why ElasticNet ties a 104M transformer: L1 sparsity matches real-response sparsity (56.8% of variance in the leading singular vector).
- Closing the gap is a training-time, not post-hoc, operation.

Limitations

- Ground truth is internally consistent but not yet externally biologically validated.
- Cross-dataset cells are single-seed; N=200 scale.

Selected references

Ahlmann-Eltze & Huber 2025, *Nat Methods*

Replogle 2022, *Cell*

Norman 2019, *Science*

Hauser & Bühlmann 2012, *JMLR* (GIES)

Brown 2025, *Nat Commun* (inspre)

Lotfollahi 2023, *Mol Syst Biol* (CPA)

Roohani 2024, *Nat Biotech* (GEARS)

Theodoris 2023, *Nature* (Geneformer)

Adduri 2025, *bioRxiv* (STATE)

Zou & Hastie 2005, *JRSS-B* (ElasticNet)